

ST AUGUSTINE PIER, FL, USA

WMO: 994410

Lat: 29.8573N

Lon: 81.2636W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.2	40.3	16.8	12.9	39.7	22.7	17.1	44.4	37.3	53.2	33.1	52.7	11.9	320	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	89.8	76.1	87.3	76.1	85.5	76.4	80.3	83.1	79.4	82.8	78.6	82.3	9.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.5	152.9	81.8	78.4	147.4	81.4	77.6	143.4	81.0	43.6	83.1	42.8	82.8	42.0	82.5	90.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
35.4	28.8	24.6	30.7	94.0	4.4	2.8	27.6	96.0	25.0	97.6	22.5	99.1	19.3	101.2
			27.3	80.9	3.4	1.8	24.8	82.2	22.8	83.3	20.8	84.3	18.3	85.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.6	55.9	58.8	62.4	68.2	74.3	78.3	79.8	80.2	78.8	73.6	65.5	59.4
	DBStd	10.25	8.01	7.41	6.73	5.09	3.76	2.74	2.63	2.47	2.54	5.39	6.78	7.99
	HDD50	78	38	15	5	0	0	0	0	0	0	0	2	18
	HDD65	927	291	194	125	28	1	0	0	0	0	10	75	203
	CDD50	7249	222	260	389	545	753	849	923	936	863	730	468	311
	CDD65	2622	10	20	43	122	289	399	458	471	413	275	91	31
	CDH74	19450	23	64	127	286	1403	3123	4095	4594	3808	1686	198	43
	CDH80	3759	1	6	19	61	212	547	962	1127	678	135	10	1
Wind	WSAvg	11.3	11.1	11.6	11.8	11.7	12.3	10.4	9.0	9.2	12.5	13.0	12.2	11.2
Precipitation	PrecAvg	50.9	2.8	2.8	3.4	2.5	3.0	6.9	5.4	6.4	8.6	4.8	2.2	2.5
	PrecMax	66.2	7.9	8.5	8.9	8.8	7.1	11.2	11.5	12.9	17.0	12.0	5.7	10.1
	PrecMin	32.9	0.9	0.6	0.2	0.6	1.0	0.7	1.7	2.9	2.0	0.3	0.2	0.4
	PrecStd	8.9	1.9	2.0	2.4	2.1	1.9	2.6	2.3	2.6	4.4	3.1	1.4	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.3	80.8	82.5	85.9	89.6	93.5	92.7	93.2	89.5	85.9	81.5	78.9
		MCWB	66.6	69.2	67.0	69.1	71.5	78.3	78.0	75.9	73.6	74.3	74.2	71.5
	2%	DB	72.0	75.3	77.7	81.1	85.8	88.0	89.1	88.7	86.1	82.9	78.1	74.3
		MCWB	64.5	65.5	65.9	68.6	71.3	76.1	77.3	77.6	74.5	74.7	73.4	68.2
	5%	DB	68.5	70.5	73.1	76.6	81.8	84.9	86.3	86.3	83.9	81.4	76.5	71.4
		MCWB	64.0	63.9	65.8	69.0	72.1	76.0	77.3	78.1	76.1	74.7	72.5	67.3
	10%	DB	66.3	67.8	70.4	74.4	79.1	82.6	84.2	84.7	82.9	80.3	74.5	69.5
		MCWB	63.1	63.6	65.2	68.6	72.6	76.1	77.4	78.0	76.1	74.0	69.5	66.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.5	70.3	70.7	73.9	77.0	79.8	83.1	81.8	80.1	80.0	78.3	71.2
		MCDB	73.3	75.2	76.0	76.5	80.3	83.9	83.3	85.5	82.6	80.5	78.9	74.5
	2%	WB	66.6	67.2	68.6	72.4	75.4	78.4	80.2	80.3	79.1	78.1	76.7	69.5
		MCDB	70.3	70.5	72.8	74.9	78.4	82.5	82.8	83.9	82.0	80.5	77.0	71.8
	5%	WB	65.2	65.0	67.4	71.2	74.6	77.6	79.0	79.5	78.1	76.5	73.3	68.3
		MCDB	67.8	67.6	70.7	73.6	78.0	81.6	82.9	83.0	81.7	79.8	75.5	70.4
	10%	WB	63.6	62.6	66.1	70.0	73.7	77.0	78.1	78.8	77.3	75.1	70.7	66.8
		MCDB	65.7	65.8	69.2	73.0	77.6	81.0	82.3	82.6	81.3	78.8	73.7	68.8
Mean Daily Temperature Range	5% DB	MDBR	12.6	11.8	11.0	10.1	8.8	9.1	9.4	8.8	7.4	8.3	10.6	11.6
		MCDBR	15.2	16.4	16.3	14.8	15.6	13.8	13.5	12.9	10.1	8.8	12.1	12.7
		MCWBR	8.6	11.2	8.6	7.3	6.2	6.1	6.2	5.7	4.4	4.8	7.5	8.9
	5% WB	MCDBR	11.0	14.1	11.0	9.5	9.0	9.6	10.0	9.2	6.9	7.0	6.1	9.2
		MCWBR	7.8	11.3	7.7	6.6	5.5	5.5	6.0	5.3	4.4	5.0	5.0	7.8
Clear-Sky Solar Irradiance	taub	0.339	0.352	0.371	0.396	0.433	0.469	0.495	0.480	0.427	0.399	0.362	0.347	
	taud	2.541	2.513	2.462	2.373	2.317	2.279	2.212	2.260	2.403	2.448	2.499	2.543	
	Ebn at Noon	285	290	291	285	273	262	255	258	269	270	273	276	
	Edh at Noon	28	31	35	39	42	43	46	43	36	33	29	26	
All-Sky Solar Radiation	RadAvg	1005	1210	1580	1928	2042	1905	1889	1770	1529	1349	1086	914	
	RadStd	74	121	135	115	131	152	115	116	124	108	64	78	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	+0.79	+1.39	N/A	N/A	N/A	N/A	-16	-112	+288	+209	

Nomenclature: See separate page